

Midterm 2 Free Response Solutions (By Version)

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Midterm 2 Free Response Solutions (Translated)

Version A Solutions

Question 22: Bob & Linda

- Absolute Advantage (Burgers):** Linda has the absolute advantage because she can produce **40 burgers** per shift, which is more than Bob's **20 burgers**.
- Bob's Opportunity Cost of 1 Burger:** Bob gives up 60 cookies to make 20 burgers. Calculation:

$$\frac{60 \text{ cookies}}{20 \text{ burgers}} = 3 \text{ cookies per burger}$$
- Comparative Advantage (Cookies):** Bob has the comparative advantage. Bob's OC of 1 Cookie is $\frac{1}{3}$ burger (20/60), while Linda's OC is $\frac{1}{2}$ burger (40/80).
- Range of Prices (Terms of Trade):** Trade will occur if the price of 1 burger is between the two workers' opportunity costs (2 and 3 cookies). Range: **Between 2 and 3 cookies per burger**.

Question 23: Pollution Permits

- Initial Total Cost:** Combined production cost for the first 6 widgets: \$10 + \$25 + \$15 + \$30 + \$20 + \$50 = **\$150**.
- WTP and WTA:**
 - Firm A Max WTP (3rd Permit):** Profit from 3rd widget = \$60 - \$40 = **\$20**.
 - Firm C Min WTA (Selling 1 Permit):** Profit lost from 2nd widget = \$60 - \$50 = **\$10**.

- c. **After-Trade Total Cost:** Cost is minimized by using the 6 permits on the lowest marginal costs available: 10, 15, 20, 25, 30, and 40. New Total Cost: **\$140**.

Version B Solutions

Question 22: Bob & Linda

- Absolute Advantage (Burgers):** Linda has the absolute advantage because she can produce **30 burgers** per shift, which is more than Bob's **10 burgers**.
- Bob's Opportunity Cost of 1 Burger:** Bob gives up 40 cookies to make 10 burgers. Calculation: $\frac{40 \text{ cookies}}{10 \text{ burgers}} = 4 \text{ cookies per burger}$
- Comparative Advantage (Cookies):** Bob has the comparative advantage. Bob's OC of 1 Cookie is $\frac{1}{4}$ burger (10/40), while Linda's OC is $\frac{1}{3}$ burger (30/90).
- Range of Prices (Terms of Trade):** Trade will occur if the price of 1 burger is between the two workers' opportunity costs (3 and 4 cookies). Range: **Between 3 and 4 cookies per burger**.

Question 23: Pollution Permits

- Initial Total Cost:** Combined production cost for the first 6 widgets: \$12 + \$20 + \$16 + \$32 + \$22 + \$48 = **\$150**.
- WTP and WTA:**
 - Firm A Max WTP (3rd Permit):** Profit from 3rd widget = \$60 - \$38 = **\$22**.
 - Firm C Min WTA (Selling 1 Permit):** Profit lost from 2nd widget = \$60 - \$48 = **\$12**.
- After-Trade Total Cost:** Cost is minimized by using the 6 permits on the lowest marginal costs available: 12, 16, 20, 22, 32, and 38. New Total Cost: **\$140**.

Version C Solutions

Question 22: Bob & Linda

- Absolute Advantage (Burgers):** Linda has the absolute advantage because she can produce **50 burgers** per shift, which is more than Bob's **25 burgers**.
- Bob's Opportunity Cost of 1 Burger:** Bob gives up 50 cookies to make 25 burgers. Calculation: $\frac{50 \text{ cookies}}{25 \text{ burgers}} = 2 \text{ cookies per burger}$
- Comparative Advantage (Cookies):** Linda has the comparative advantage. Linda's OC of 1 Cookie is $\frac{1}{3}$ burger (50/150), while Bob's OC is $\frac{1}{2}$ burger (25/50).
- Range of Prices (Terms of Trade):** Trade will occur if the price of 1 burger is between the two workers' opportunity costs (2 and 3 cookies). Range: **Between 2 and 3 cookies per burger**.

Question 23: Pollution Permits

- Initial Total Cost:** Combined production cost for the first 6 widgets: \$15 + \$30 + \$20 + \$35 + \$25 + \$60 = **\$185**.
- WTP and WTA:**
 - Firm A Max WTP (3rd Permit):** Profit from 3rd widget = \$70 - \$45 = **\$25**.
 - Firm C Min WTA (Selling 1 Permit):** Profit lost from 2nd widget = \$70 - \$60 = **\$10**.
- After-Trade Total Cost:** Cost is minimized by using the 6 permits on the lowest marginal costs available: 15, 20, 25, 30, 35, and 45. New Total Cost: **\$170**.

Version D Solutions

Question 22: Bob & Linda

- Absolute Advantage (Burgers):** Linda has the absolute advantage because she can produce **20 burgers** per shift, which is more than Bob's **15 burgers**.
- Bob's Opportunity Cost of 1 Burger:** Bob gives up 60 cookies to make 15 burgers. Calculation:
 $\frac{60 \text{ cookies}}{15 \text{ burgers}} = 4 \text{ cookies per burger}$
- Comparative Advantage (Cookies):** Bob has the comparative advantage. Bob's OC of 1 Cookie is $\frac{1}{4}$ burger (15/60), while Linda's OC is $\frac{1}{2}$ burger (20/40).
- Range of Prices (Terms of Trade):** Trade will occur if the price of 1 burger is between the two workers' opportunity costs (2 and 4 cookies). Range: **Between 2 and 4 cookies per burger**.

Question 23: Pollution Permits

- Initial Total Cost:** Combined production cost for the first 6 widgets: \$20 + \$30 + \$25 + \$35 + \$30 + \$65 = **\$205**.
- WTP and WTA:**
 - Firm A Max WTP (3rd Permit):** Profit from 3rd widget = \$80 - \$40 = **\$40**.
 - Firm C Min WTA (Selling 1 Permit):** Profit lost from 2nd widget = \$80 - \$65 = **\$15**.
- After-Trade Total Cost:** Cost is minimized by using the 6 permits on the lowest marginal costs available: 20, 25, 30, 30, 35, and 40. New Total Cost: **\$180**.

Version E Solutions

Question 22: Bob & Linda

- Absolute Advantage (Burgers):** Linda has the absolute advantage because she can produce **60 burgers** per shift, which is more than Bob's **30 burgers**.
- Bob's Opportunity Cost of 1 Burger:** Bob gives up 90 cookies to make 30 burgers. Calculation:
 $\frac{90 \text{ cookies}}{30 \text{ burgers}} = 3 \text{ cookies per burger}$
- Comparative Advantage (Cookies):** Bob has the comparative advantage. Bob's OC of 1 Cookie is $\frac{1}{3}$ burger (30/90), while Linda's OC is $\frac{1}{2}$ burger (60/120).
- Range of Prices (Terms of Trade):** Trade will occur if the price of 1 burger is between the two workers' opportunity costs (2 and 3 cookies). Range: **Between 2 and 3 cookies per burger**.

Question 23: Pollution Permits

- Initial Total Cost:** Combined production cost for the first 6 widgets: \$5 + \$15 + \$10 + \$20 + \$15 + \$40 = **\$105**.
- WTP and WTA:**
 - Firm A Max WTP (3rd Permit):** Profit from 3rd widget = \$50 - \$25 = **\$25**.
 - Firm C Min WTA (Selling 1 Permit):** Profit lost from 2nd widget = \$50 - \$40 = **\$10**.
- After-Trade Total Cost:** Cost is minimized by using the 6 permits on the lowest marginal costs available: 5, 10, 15, 15, 20, and 25. New Total Cost: **\$90**.